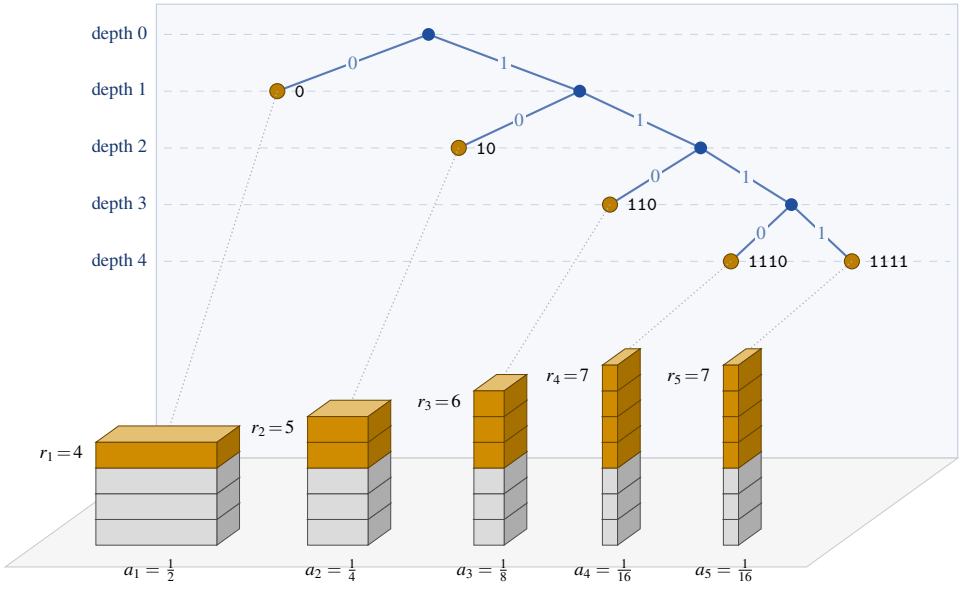


shift $k = 3$, budget $\theta = 2^{-3}$

code length ℓ_i of leaf i

$r_i = \ell_i + k$

$$\sum_i 2^{-r_i} = 2^{-k} \sum_i 2^{-\ell_i} = 2^{-k}$$

volume $\sum_i a_i r_i = A\left(k + \bar{\ell}\right)$

$\bar{\ell} = H_2(p) = 1.875$ bits